

# GOGADA PRAMEELA

Vizianagaram, Andhra Pradesh | +91 70938 11130 | [prameelagogada4@gmail.com](mailto:prameelagogada4@gmail.com)

## SUMMARY

---

Final-year B.Tech Electronics and Communication Engineering student (2027) building an automotive CAN-bus validation project with Raspberry Pi, ESP32 and Python. Foundation in C and Python, 98% in SSC. Seeking an entry-level role in embedded systems, hardware validation or software engineering.

## EDUCATION

---

|                                                          |                 |
|----------------------------------------------------------|-----------------|
| <b>B.Tech, Electronics and Communication Engineering</b> | 2027 (expected) |
| Avanthi Institute of Engineering & Technology            | 75%             |
| <b>Intermediate (12th)</b>                               | 2022            |
| Punyagiri Junior College                                 | 65%             |
| <b>SSC (10th)</b>                                        | 2020            |
| Z.P.H. School, Jami                                      | 98%             |

## ACADEMIC PROJECT

---

**Automated Validation and Anomaly Detection for In-Vehicle CAN Networks** Sep 2026 – Apr 2027

*Final-year project (in progress)*

**Technologies:** Embedded C, Python, Linux, Raspberry Pi 4, ESP32-S3, CAN (SocketCAN, DBC), pytest

- Building a three-node CAN network on the bench: a Raspberry Pi 4 gateway and two ESP32-S3 boards acting as engine and ABS control units (ECUs).
- Developing a C program on Linux SocketCAN that decodes vehicle signals (speed, RPM, temperature) from a DBC file and flags out-of-range values.
- Creating an automated pytest test suite with fault injection (invalid values, missing messages) that generates PASS/FAIL reports.
- Next phases: machine-learning detection of spoofed CAN messages, and decoding a 433 MHz tyre-pressure sensor with an RTL-SDR.

## TECHNICAL SKILLS

---

- **Programming:** C, Python, MATLAB
- **Embedded and tools (learning through project):** Raspberry Pi, ESP32, Linux, CAN, UART, SPI, I<sup>2</sup>C, Git
- **AI-assisted development (vibe coding):** building and debugging web pages and scripts with AI coding tools (Claude, GitHub Copilot)
- **Relevant coursework:** Digital Electronics, Microprocessors and Microcontrollers, Signals and Systems, Communication Systems

## LANGUAGES AND INTERESTS

---

- **Languages:** English, Telugu, Hindi
- **Interests:** Space technology, exploring new technologies